A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 21

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (120 + 95) - (2 + 45) = 168 t ha^{-1} **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (t ha^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO2 **1** Harvest removes biomass so less dead organic matter is returned – pasture roots and litter constantly add organic matter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Low / zero tillage **1** Adding manure, compost or mulch **1** Growing cover crops / leaving crop residues **1** Converting to long-term grass or tree crops in rotation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed – hydrodynamics for ships **2** Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption **2** Conventional friction brakes convert kinetic energy to waste heat – regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Pneumatic tyres deform as they rotate – the continual change in shape and air movement inside generates frictional heat (more if under-inflated) **1** Steel wheels do not deform so rolling resistance is very low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel **1** Better temperature control (thermostatic fan) **1** Electronic ignition control / regular servicing to keep spark timing correct **1** Turbochargers supplying more oxygen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1
03.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (150 - 125) / 150 x 100 = 16.7% (accept 17%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed fee gives no incentive to save water because the bill is the same however much is used **1** Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Low-water appliances:** dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers **1 Greywater re-use:** bath / shower water used to flush toilets or water gardens **1** Rainwater collection / water butts for garden use **1 Xeriscaping:** planting drought-adapted species that need no watering **1** Repairing leaks / dripping taps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Old infrastructure / pipes in poor condition leak **1** Repair and good maintenance / pipe replacement / leak detection reduce wastage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.5	**1 mark each** 1 mark for each of the 2 points below. **1** Water is delivered directly to the plant roots / soil around each plant **1** Less evaporation loss from spray droplets and from wetting the whole soil surface – less runoff **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1 Gravimetry:** variations in gravity caused by differences in the density and mass of the rock below – ore-bearing rocks are usually denser than the surrounding sedimentary deposits **1 Magnetometry:** rocks that are more magnetic than their surroundings, such as the iron ore magnetite and the ores of tungsten and cobalt **1 Seismic surveys:** echoes of sound waves from controlled explosions or a surface vibrator, giving the depth, density and shape of the rock strata **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trial drilling is the most expensive technique per sampling site, so the remote surveys are used first to narrow the search to the most promising ground **1** Drilling is the only technique that brings up actual samples of the rock, so it is used to confirm the composition and purity found by chemical analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reserves are the deposits that can be extracted economically with current technology – the resource base is everything that exists, including deposits that are too low-grade, too deep or too remote to pay **1** A higher price lowers the cut-off ore grade, so deposits that previously could not be mined at a profit become worth extracting and are reclassified as reserves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** All the rock above the ore (the overburden) must be stripped off, so habitat is destroyed over a much larger area than with deep mining **1** Spoil heaps of waste rock occupy land, may be unstable and can be colonised only slowly because they hold few nutrients and little water **1** Dust from blasting, crushing and haulage raises atmospheric particulate levels and settles on surrounding vegetation **1** Noise and vibration from blasting and heavy machinery disturbs wildlife and nearby residents **1** Drainage from the workings and spoil is often acidic and carries dissolved metals, polluting watercourses and groundwater **1** Lowering the water table to keep the pit dry dries out nearby wetlands and wells **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
05.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Topsoil and subsoil stripped at the start are stored separately and replaced in the correct order, so the restored soil has a fertile surface layer **1** The land is re-contoured and spoil heaps are given shallow, stable slopes before being covered with soil and planted **1** The site can be returned to agriculture or forestry, or flooded workings developed as a wetland or water-sport habitat **1** Planning restoration first means the topsoil is saved rather than buried, the final landform is built as mining proceeds, and the cost is included in the economic assessment of the mine **1** A restoration bond or condition in the planning permission makes certain the work is paid for even if the company fails **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 136-142; AQA spec 3.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic **1** Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it – the cold water returns southwards at depth (thermohaline circulation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** Higher temperatures melt land ice on Greenland which flows into the North Atlantic **1** The fresh water dilutes the seawater, reducing its salinity and density **1** Less dense water sinks less easily so the flow rate of the current is reduced **1** Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them **1** Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change – individual events do not show trends **1** Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity) – incomplete understanding of the processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anthropogenic** → human-caused / caused by human activity	3
06.4	1 mark for the point below. **1** Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data – the sequence of positions shows current direction and speed (surface currents also by satellites and buoys) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora **1** 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants) – 100 m x 100 m is for trees (impractical, too much work) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
07.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = 240 – N(N-1) = 240 x 239 = 57 360 **1** Σn(n-1) = 48x47 + 120x119 + 36x35 + 24x23 + 12x11 = 2256 + 14280 + 1260 + 552 + 132 = 18 480 **1** D = 57 360 / 18 480 = 3.10 (accept 3.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance • **Biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
07.3	**1 mark each** 1 mark for each of the 2 points below. **1 Lower D means lower biodiversity:** the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) **1** Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **plagioclimax** → a community held back from its climax by human activity	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants **1** Point quadrat pins are lowered through the vegetation – each touch counts (10 pins = 10% each) without flattening **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
07.5	1 mark for the point below. **1** Subjective judgement varies between recorders – results are not quantitative so statistical analysis is not possible (but quick for many areas) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Landfill fell from 79% to 14% (large decrease) **1** Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Incineration** → burning waste, usually with energy recovery • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Organic matter decays anaerobically releasing methane (and CO2) - greenhouse gases **1** Toxic leachate may leak into groundwater / rivers from poorly managed sites **1** Uses large areas of land – loss of habitats / farmland – contamination prevents later use **1** Resource value of metals, glass, plastics is lost **1** Transport noise / congestion / litter / pests / odour **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
08.3	**Any 4 of 7** 1 mark each for any 4 of the 7 points below, in any order, maximum 4. **1** Polymer / clay liner to prevent escape of leachate **1** Collection and treatment of leachate **1** Collection of methane and use as a fuel **1** Impermeable cap (clay and polymer) when complete to stop water entering **1** Regular covering with soil (pests, odour) – deodorising sprays **1** Perimeter fence to trap wind-blown litter **1** Separation and recording of waste types – reactive / flammable wastes not dumped together **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste • **Impermeable** → does not let water through **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	4
08.4	1 mark for the point below. **1** Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	
09.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Organic (methyl) mercury **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Anaerobic microbes converted the inorganic mercury into organic methyl mercury **1** Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are less soluble at high pH **1** So they remain immobile and are not leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
10.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 4.2 / 0.7 = 6 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low **1** In winter days are very short / the Sun does not rise, so light levels are extremely low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reflectors can only focus parallel rays of direct sunlight onto the absorber **1** Clouds, smoke and dust scatter the light so it cannot be focused and intensified **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.4	1 mark for the point below. **1** Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
10.5	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Problems:** intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • **Photothermal:** evacuated tubes reduce conduction / convection losses – thermal stores (water, sand, concrete, molten salt) allow later use – passive solar architecture, brise soleil, light tubes – heat pumps deliver up to four times the energy used • **Photovoltaic developments:** multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • **Integration:** grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • **Evaluation:** strong in sunny deserts / low latitudes, weaker in high latitudes in winter – cost falling – manufacturing impacts – no single technology solves all problems – justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **pumped storage** → water pumped uphill when demand is low and released to generate at peak **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (9 marks)	
	Solar power's problems are intrinsic: it is intermittent with daily and seasonal cycles, cloud makes it unpredictable and scatters light so it cannot be focused, its low energy density demands large collector areas, the angle of incidence keeps changing, cleaning in deserts uses scarce water, manufacturing PV produces toxic wastes such as silicon tetrachloride and cadmium, and PV efficiency falls as cells heat up.	
	New technology addresses most of these. Evacuated-tube photothermal panels cut conduction and convection losses, and thermal stores of water, sand, concrete or molten salt hold heat for later; passive solar design, brise soleil and light tubes reduce demand; heat pumps deliver up to four times the energy used to run them. Multijunction PV cells absorb more wavelengths, moth-eye anti-reflective surfaces stop the 30% of light a smooth cell reflects, PV-thermal hybrids cool the cell and use the heat, transparent PV turns windows into generators, heliostats keep panels at the optimum angle and nanohydrophobic surfaces self-clean. Concentrating solar power with molten salt stored at 550 °C runs steam turbines when electricity is wanted, directly overcoming intermittency, and grid-connected farms, rooftops that avoid land conflicts, batteries and pumped storage integrate the output.	
	Even so, no technology changes the fact that winter insolation at high latitudes is tiny, that CSP needs cloudless skies, or that manufacturing has impacts. Solar can dominate supply in sunny, low-latitude regions and contribute strongly elsewhere in summer, but overcoming its problems fully depends on storage and a diverse energy mix rather than on solar technology alone.	
	Weaker answer - Level 1 (2 marks)	
	Solar power has problems because it does not work at night or when it is cloudy, and solar panels are not very efficient. New technologies can help. Better solar panels are more efficient. Solar power can be stored in batteries for use at night. Mirrors can concentrate the sunlight to make it hotter. So new technologies can overcome most of the problems with solar power.	
	What makes the difference	
	• Level 1 identifies two problems and three vague technologies; Level 3 names each technology and explains which problem it solves and how.	
	• Level 3 acknowledges what technology cannot fix (high-latitude winters, cloud for CSP, manufacturing impacts) - the 'extent' in the question.	
	• Numbers (30% reflection, 550 °C, four-fold heat-pump output) show detailed knowledge applied to the argument.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • **Fossil fuels:** choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports – new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • **Nuclear:** high energy density reduces transport – waste properties (half-life) determine storage – thorium and fusion prospects • **Renewables:** matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting) – intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity – low-impact designs (helical turbines, VAWT, tidal lagoons) • **Energy conservation reduces demand:** transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use	25
	– life-cycle / embodied energy • **Evaluation:** no resource is impact-free – knowledge allows the best mix and mitigation – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

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